Pravin Anbalagan

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EXECUTIVE SUMMARY

Computer Science undergraduate at VIT Vellore with proven experience in full-stack development, database-driven applications, and deep learning systems. Built end-to-end platforms integrating REST APIs, SQL databases, and AI models — from agricultural marketplace systems to GPT-style language models. Strong foundation in data structures, algorithms, and object-oriented programming. Actively pursuing AI/ML engineering roles to apply research-level understanding to real-world problems.

WORK EXPERIENCE

Software Engineering Intern – AMMAN TRY *2026*

- Managed and structured customer interaction data within CRM software, ensuring data accuracy, consistency, and integrity across records.
- Executed and tracked telephonic engagement workflows through the CRM platform, using system-based logging and status updates to streamline customer outreach.
- Updated and maintained customer records through CRM data-entry tools, applying structured field validation to keep the database reliable.
- Utilised CRM system features — record tagging, status pipelines, and reporting views — to support customer lifecycle tracking and improve engagement data analysis.

Open Source Contributor | Social Summer of Code (SSoC) *2026*

- Developed and merged features, resolved issues, and collaborated with maintainers using GitHub pull requests, code reviews, and version control.
- Developed an end-to-end NLP pipeline for analyzing large-scale product reviews using DistilBERT, BERTopic, SentenceTransformers, TF-IDF, and Logistic Regression to perform sentiment analysis, topic modeling, and aspect-based sentiment extraction.
- Generated actionable business insights by identifying market gaps, product comparison metrics, ingredient analytics, authenticity flags, and trend reports through automated data processing and visualization.

PROJECTS

Agrow – Agricultural Marketplace Platform *2025*

Full Stack | Python, SQL, REST APIs, Blockchain

- Designed and deployed a full-stack marketplace connecting 50+ simulated farmers, buyers, and logistics providers; eliminated middlemen enabling direct trade with live market price feeds via REST API integration.
- Engineered a blockchain-based batch identification system achieving 100% crop traceability across 3 supply chain tiers; reduced simulated transaction fraud by ~40% through smart-contract escrow releasing funds only on delivery confirmation.
- Optimised location-based order grouping algorithm (Python + SQL) cutting simulated transport cost by 25%; system handled 200+ concurrent order records with sub-200ms query response.

FootballGPT – Transformer Language Model from Scratch *2026*

Python, PyTorch | Trained on 15 MB World Cup dataset

- Implemented a full Transformer architecture (nanoGPT) from scratch — tokenizer, multi-head self-attention, positional encoding, residual connections, layer normalisation — without high-level ML libraries; 6-layer, 10M parameter model.
- Trained on a 15 MB football dataset achieving cross-entropy loss of 1.82 after 5,000 iterations; model generates contextually coherent football commentary and Q&A via autoregressive decoding.
- Gained deep expertise in Query-Key-Value attention mechanisms and transformer block stacking — core concepts underlying production LLMs such as GPT-4.

Deep3DFaceRecon – 3D Face Reconstruction from a Single Image *2026*

Python, PyTorch, OpenCV, Open3D, ResNet-50, Computer Vision

- Implemented a weakly supervised CNN-based 3D face reconstruction pipeline (ResNet-50 + BFM 2019, 53K-vertex mesh) to reconstruct textured facial geometry from single RGB images, integrating landmark alignment, photometric refinement, texture transfer, and OBJ mesh export.
- Diagnosed and resolved rendering artifacts through surface-normal–based eye socket correction and depth-based vertex classification, and extended the pipeline with an Open3D interactive viewer for real-time 3D mesh visualization.

EDUCATION

B.Tech Computer Science Engineering | CGPA: 8.68 / 10 *2024 – 2028*

Vellore Institute of Technology, Vellore

TECHNICAL SKILLS

Languages: Python, C, C++, Java, R
ML/AI Libraries: TensorFlow, OpenCV, NumPy, Pandas, PyTorch
Tools & Databases: Git, Jupyter Notebook, SQLite, SQL, PL/SQL basics, Streamlit, REST APIs
Core Concepts: Machine Learning, Deep Learning, Transformers, Data Structures & Algorithms, OOP, Dynamic Programming

CERTIFICATIONS

- Machine Learning Specialization – DeepLearning.AI (Coursera)
- Generative AI for Everyone – DeepLearning.AI (Coursera)
- Generative AI in Business – University of Michigan